RADIATOR (Continued)

## INSTALLATION

**CAUTION: Before installing the radiator or A/C condenser, be sure the radiator-to-body and radiator-to-A/C condenser rubber air seals are properly fastened to their original positions. These are used at the top, bottom and sides of the radiator and A/C condenser. To prevent overheating, these seals must be installed to their original positions.**

(1) Guide the two radiator alignment dowels into the rubber grommets located in lower radiator crossmember. Install and tighten the six mounting bolts (Fig. 23) to 8 N·m (72 in. lbs.) torque.
(2) Close radiator draincock.
(3) 2.4L ONLY - Install the electric cooling fan (Refer to 7 - COOLING/ENGINE/RADIATOR FAN - INSTALLATION).
(4) Position fan shroud and power steering reservoir tank (if equipped). Install and tighten four mounting bolts to 8 N·m (72 in. lbs.) torque.
(5) If equipped, remove plugs and connect automatic transmission fluid cooler lines and constant tension clamps.

**CAUTION: The tangs on the hose clamps must be positioned straight down.**

(6) Connect radiator hoses and install hose clamps.
(7) Position and install the condenser lower seal (Fig. 22).
(8) Connect battery negative cable.
(9) Fill cooling system with correct coolant (Refer to 7 - COOLING - STANDARD PROCEDURE).
(10) Connect coolant recovery bottle hose.
(11) Install radiator cap.
(12) Check and adjust automatic transmission fluid level (if equipped).
(13) Start engine and check for leaks.

# WATER PUMP - 2.4L

## DESCRIPTION

The water pump has a cast aluminum body and housing with a stamped steel impeller. The water pump bolts directly to the block (Fig. 25). The cylinder block to water pump seal is provided by a rubber O-ring. The water pump is driven by the engine timing belt.

## OPERATION

The water pump is the heart of the cooling system. The coolant is pumped through the engine block, cylinder head, heater core, and radiator.

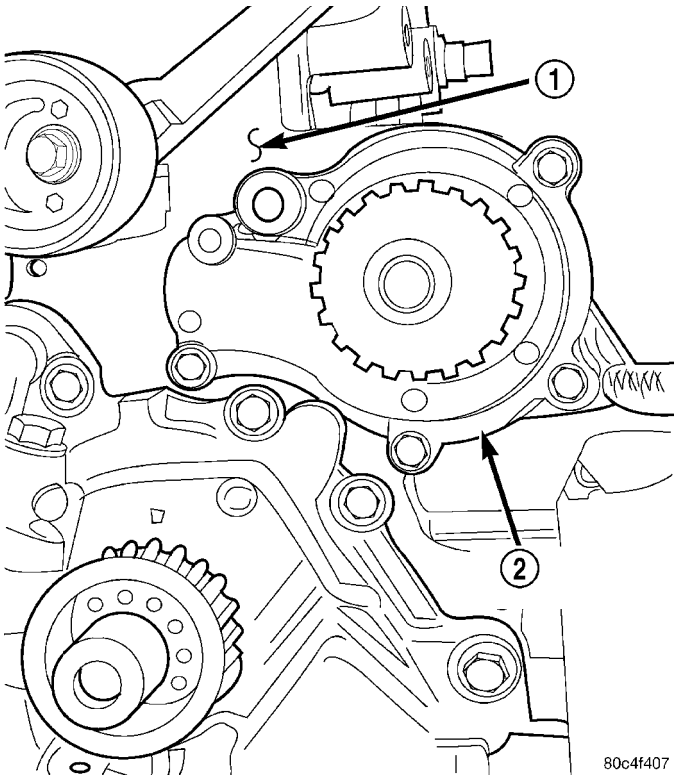


***Fig. 25 Water Pump***

1 - CYLINDER BLOCK
2 - WATER PUMP

# DIAGNOSIS AND TESTING

## WATER PUMP

A quick flow test to determine if the water pump is working effectively is to check heater system for proper operation. A defective pump will not provide an adequate flow of heated coolant through the system.

**WARNING: DO NOT REMOVE THE COOLING SYSTEM PRESSURE CAP WITH THE SYSTEM HOT AND UNDER PRESSURE BECAUSE SERIOUS BURNS FROM COOLANT CAN OCCUR.**

Another flow test to help determine water pump operation is as follows:
(1) Remove cooling system pressure cap.
(2) Remove a small amount of coolant from the system.
(3) Start the engine and warm up until thermostat opens.
(4) With the thermostat open and coolant level low, visually inspect for coolant flow. If flow is present, the water pump is pumping coolant through the system.